Circles

The big idea

A circle is the set of all points at a fixed distance (radius) from a centre. Chords, arcs and the angles they make follow beautiful rules that let us prove many geometry results.

Chords and their distances

CHORD THEOREMS

Equal chords subtend **equal angles** at the centre (and conversely).
The **perpendicular from the centre** to a chord **bisects** the chord; the line from the centre to the midpoint of a chord is perpendicular to it.
There is **one and only one circle** through three non-collinear points.
Equal chords are equidistant from the centre (and conversely).

A CHORD OF LENGTH 8 CM IS 3 CM FROM THE CENTRE. FIND THE RADIUS.

The perpendicular bisects the chord, so the half-chord is 4 cm. $r = \sqrt{(4^2 + 3^2)} = \mathbf{5\ cm}$.

Angles subtended by an arc

ANGLE THEOREMS

The angle an arc subtends at the **centre** is **double** the angle it subtends at any point on the remaining part of the circle.

Angles in the same segment are equal.

The angle in a **semicircle** is a right angle (90°).

AN ARC SUBTENDS 100° AT THE CENTRE. WHAT ANGLE DOES IT SUBTEND AT THE CIRCLE?

Half of 100° = **50°**.

Cyclic quadrilaterals

A quadrilateral whose four vertices lie on a circle is **cyclic**.

CYCLIC QUADRILATERAL THEOREM

Opposite angles of a cyclic quadrilateral are **supplementary** (sum 180°). Conversely, if a quadrilateral's opposite angles sum to 180°, it is cyclic.

ABCD IS CYCLIC WITH ∠A = 75°. FIND ∠C.

∠C = 180° − 75° = **105°**.

Quick recap

- Perpendicular from the centre bisects a chord.
- Equal chords: equal central angles, equal distances from the centre.
- Central angle = 2 × angle at the circle; angle in a semicircle = 90°.
- Opposite angles of a cyclic quadrilateral sum to 180°.

Practice questions

1. The angle in a semicircle is
   (a) 45° (b) 60° (c) 90° (d) 180°
2. A chord 6 cm long is 4 cm from the centre. The radius is
   (a) 4 cm (b) 5 cm (c) 6 cm (d) 10 cm
3. An arc subtends 70° at a point on the circle. At the centre it subtends
   (a) 35° (b) 70° (c) 140° (d) 110°
4. In a cyclic quadrilateral ABCD, $\angle B = 110°$. Then $\angle D$ is
   (a) 70° (b) 110° (c) 90° (d) 250°
5. How many circles pass through three non-collinear points?
   (a) None (b) Exactly one (c) Two (d) Infinitely many

Answers: 1. c 2. b 3. c 4. a 5. b